- pre calculated values for $\gamma = 1.4$

AA 283: Aircraft and Rocket Propulsion

Compressible Flow Equations

Spring 2026

1 Isentropic Flow Equations

These can be found on NASA's isentropic flow page.

The Mach number (1) is defined as the ratio between the velocity and the speed of sound. The speed of sound is dependent on the medium and the state of the medium, and is defined as Eq. (2).

$$M = \frac{u}{a} \tag{1}$$

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} = \sqrt{\gamma R T} \tag{2}$$

remember to use STATIC temperature here

The total conditions are defined as the conditions of the flow after it has been brought to rest. They are also sometimes referred to as the *stagnation* conditions. The pressure, density, and temperature ratios can be inferred from one another using the equation of state, as seen by Eq. (4).

$$\frac{p}{\rho^\gamma} = \text{const.} = \frac{p_t}{\rho_t^\gamma} \tag{3}$$

$$\frac{p}{p_t} = \left(\frac{\rho}{\rho_t}\right)^\gamma = \left(\frac{T}{T_t}\right)^{\frac{\gamma}{\gamma-1}} \tag{4}$$

3.5

The dynamic pressure q is a function of the static density and current velocity.

$P = \rho R T$ (ideal gas)

$$q = \frac{1}{2}\rho u^2 = \frac{\gamma}{2} p M^2 \tag{5}$$

The total pressure, total temperature, and total density ratios are all derived from the conservation equations.

$$\frac{p_t}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}} \tag{6}$$

3.5

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2 \tag{7}$$

$$\frac{\rho_t}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{\gamma-1}} \tag{8}$$

2.5

The choked area ratio (9) is very useful for nozzle calculations. It is derived from the compressible mass flow equation (10). The derivation of the mass flow equation can be found in AA283_Supplementary_005_Mass_Flow.pdf.

$$f(M) = \frac{A^*}{A} = \frac{\gamma+1}{2}^{\overset{3}{\left(\frac{\gamma+1}{2(\gamma-1)}\right)}} \frac{M}{\left(1+\frac{\gamma-1}{2}M^2\right)^{\underset{3}{\frac{\gamma+1}{2(\gamma-1)}}}}$$

$$f^{-1}(M) = \frac{A}{A^*} = \overset{1.2}{\frac{\gamma+1}{2}}^{\overset{-3}{-\frac{\gamma+1}{2(\gamma-1)}}} \frac{\left(1+\overset{0.2}{\frac{\gamma-1}{2}}M^2\right)^{\overset{3}{\frac{\gamma+1}{2(\gamma-1)}}}}{M} \tag{9}$$

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}}\sqrt{\frac{\gamma}{R}}M\left(1+\overset{0.2}{\frac{\gamma-1}{2}}M^2\right)^{\overset{-3}{-\frac{\gamma+1}{2(\gamma-1)}}} \tag{10}$$

When a flow is choked, the Mach number reaches unity. From here, the choked flow equation is produced (11).

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}}\overset{0.0698}{\sqrt{\frac{\gamma}{R}}}\left(1+\frac{\gamma-1}{2}\right)^{\overset{-3}{-\frac{\gamma+1}{2(\gamma-1)}}} \tag{11}$$

There is a critical pressure ratio formula that determines if the flow is choked or not. In general, if the reservoir pressure is at least two times higher than the backpressure, the flow will be choked. The critical pressure ratio is dependent solely on γ.

$$\frac{p_t}{p_b} \geq \underset{1.2}{\left(\frac{\gamma+1}{2}\right)}^{\overset{3.5}{\frac{\gamma}{\gamma-1}}} \tag{12}$$

2 Rayleigh Flow

The star (*) condition represents the choking (or critical or sonic) condition, at which the Mach number reaches unity. By substituting in $M=1$, the ratios for all conditions with respect to the critical condition can be produced.

Equations bypassing * ref. values

$$\frac{p_2}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}$$

$$\frac{T_2}{T_1} = \left(\frac{M_2(1+\gamma M_1^2)}{M_1(1+\gamma M_2^2)}\right)^2 = \left(\frac{M_1 p_2}{M_2 p_1}\right)^2$$

$$\frac{T_{t2}}{T_{t1}} = \frac{T_2}{T_1}\left(\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right)$$

$$\frac{p_{t2}}{p_{t1}} = \frac{p_2}{p_1}\left(\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{p}{p^*} = \frac{\gamma+1}{1+\gamma M^2} \tag{13}$$

$$\frac{T}{T^*} = M^2\left(\frac{\gamma+1}{1+\gamma M^2}\right)^2 \tag{14}$$

$$\frac{\rho}{\rho^*} = \frac{1}{M^2}\left(\frac{1+\gamma M^2}{\gamma+1}\right) \tag{15}$$

$$\frac{T_t}{T_t^*} = \underbrace{M^2\left(\frac{\gamma+1}{1+\gamma M^2}\right)^2}_{T/T^*}\left(\frac{1+\overset{0.2}{\frac{\gamma-1}{2}}M^2}{\underset{1.2}{\frac{\gamma+1}{2}}}\right) \tag{16}$$

$$\frac{p_t}{p_t^*} = \underbrace{\left(\frac{\gamma+1}{1+\gamma M^2}\right)}_{p/p^*}\left(\frac{1+\overset{0.2}{\frac{\gamma-1}{2}}M^2}{\underset{1.2}{\frac{\gamma+1}{2}}}\right)^{\overset{3.5}{\frac{\gamma}{\gamma-1}}} \tag{17}$$

The Gibbs equation for entropy is written as Eq. (18a). With the inclusion of the isentropic flow relations, the temperature ratio (T^*/T) and pressure ratio (p^*/p) can be substituted to produce Eq. (18b), a function with respect to the Mach number.

$$s^* - s = c_p \ln\left(\frac{T^*}{T}\right) - R\ln\left(\frac{p^*}{p}\right) \tag{18a}$$

$$\frac{s-s^*}{c_p} = \ln\left[M^2\left(\frac{\gamma+1}{1+\gamma M^2}\right)^{\frac{\gamma+1}{\gamma}}\right] \tag{18b}$$

The derivation of the enthalpy equation requires the assumption of a calorically perfect gas, such that Eq. (19a) holds true. Once again, with the isentropic flow equations, the final enthalpy equation (19b) is derived.

$$h = c_p T \tag{19a}$$

$$\frac{h}{h^*} = \frac{(\gamma+1)^2 M^2}{(1+\gamma M^2)^2} \tag{19b}$$

Entropy in reference to an initial condition:

$$\frac{s-s_1}{c_p} = \ln\frac{T}{T_1} - \underbrace{\frac{\gamma-1}{\gamma}}_{2/7}\ln\left[\frac{(1+\gamma M_1^2) \pm \sqrt{(1+\gamma M_1^2)^2 - 4\gamma M_1^2 (T/T_1)}}{2}\right] \tag{20}$$

3 Rankine-Hugoniot Shock Jump

The Rankine-Hugoniot Shock Jump equations are used for normal shocks. They assume that the shock is adiabatic (no change in total temperature) and that the conservations equations are all held.

$$M_2^2 = \frac{2+(\gamma-1)M_1^2}{2\gamma M_1^2 - (\gamma-1)} \tag{21a}$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2} \tag{21b}$$

$$\frac{p_2}{p_1} = 1 + \overset{7/6}{\frac{2\gamma}{\gamma+1}}(M_1^2-1) \tag{21c}$$

$$\frac{T_2}{T_1} = \left[1 + \overset{7/6}{\frac{2\gamma}{\gamma+1}}(M_1^2-1)\right]\left[\frac{2+(\gamma-1)M_1^2}{(\gamma+1)M_1^2}\right] \tag{21d}$$

$$\frac{p_{t2}}{p_{t1}} = \left[1 + \overset{7/6}{\frac{2\gamma}{\gamma+1}}(M_1^2-1)\right]\left[\frac{1+\overset{0.2}{\frac{\gamma-1}{2}}\left(\frac{2+(\gamma-1)M_1^2}{2\gamma M_1^2-(\gamma-1)}\right)}{1+\underset{0.2}{\frac{\gamma-1}{2}}M_1^2}\right]^{\frac{\gamma}{\gamma-1}\ 3.5} \tag{21e}$$

$$\Delta s/R = -\ln p_{t2}/p_{t1} \tag{21f}$$

* Note, you CANNOT use these directly on total values.

The workflow across a shock should go

$$\begin{matrix} T_{t1}, P_{t1} \\ M_1 \end{matrix} \xrightarrow[\text{isentropic}]{} \begin{matrix} T_1, P_1 \\ M_1 \end{matrix} \xrightarrow[\text{RH normal}]{} \begin{matrix} T_2, P_2 \\ M_2 \end{matrix} \xrightarrow[\text{isentropic}]{} \begin{matrix} T_{t2}, P_{t2} \\ M_2 \end{matrix}$$

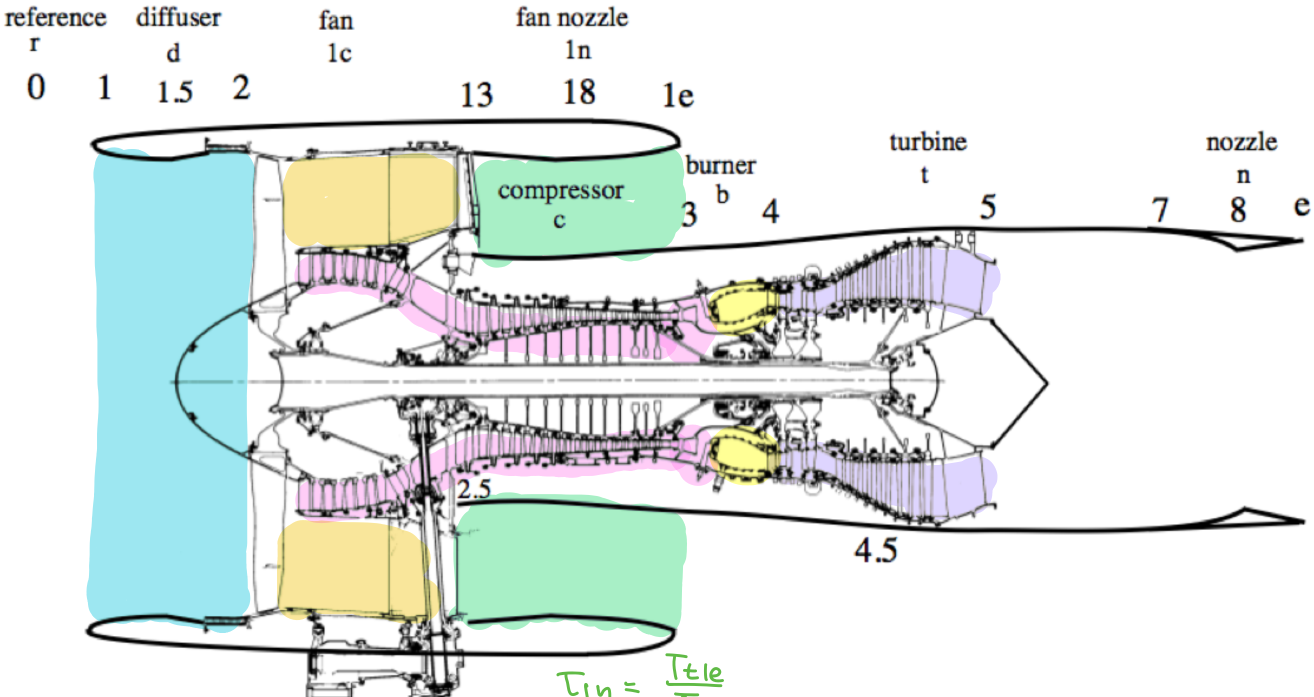
FANNO FLOW (just in case for scramjets)

$$\frac{P_2}{P_1} = \frac{M_1}{M_2}\sqrt{\frac{2+(\gamma-1)M_1^2}{2+(\gamma-1)M_2^2}} = \frac{M_1}{M_2}\sqrt{\frac{T_2}{T_1}}$$

$$\frac{T_2}{T_1} = \frac{2+(\gamma-1)M_1^2}{2+(\gamma-1)M_2^2}$$

$$\frac{P_{t2}}{P_{t1}} = \frac{M_1}{M_2}\left[\frac{2+(\gamma-1)M_2^2}{2+(\gamma-1)M_1^2}\right]^{\overset{=3}{\frac{\gamma+1}{2(\gamma-1)}}} = \frac{M_1}{M_2}\left(\frac{T_2}{T_1}\right)^{\overset{=3}{\frac{\gamma+1}{2(\gamma-1)}}}$$

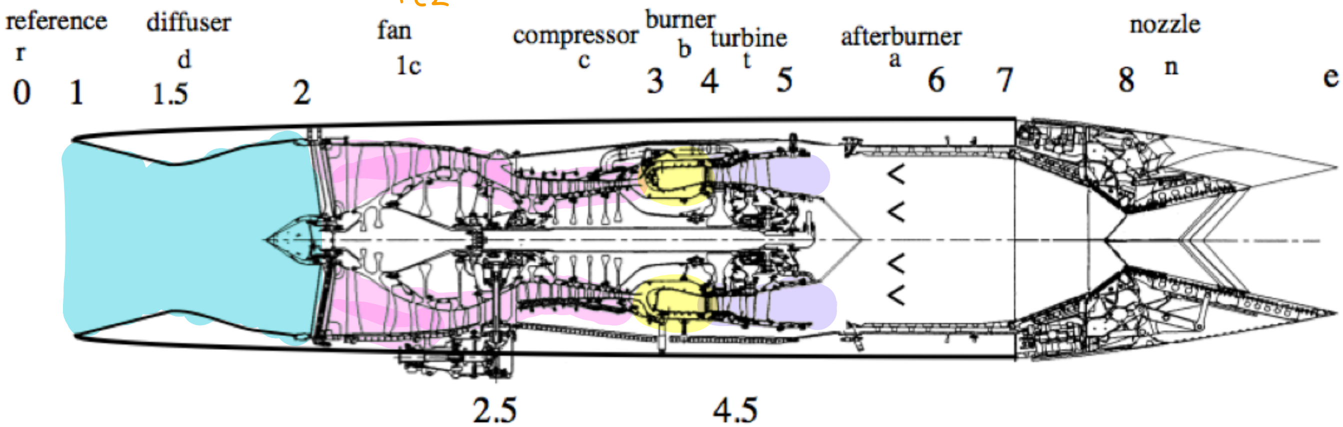
Commercial Turbofan



$$\tau_{1c} = \frac{T_{t13}}{T_{t2}} \qquad \pi_{1c} = \frac{P_{t13}}{P_{t2}}$$

$$\tau_{1n} = \frac{T_{t1e}}{T_{t13}} \qquad \pi_{1n} = \frac{P_{t1e}}{P_{t13}}$$

Military Turbofan



$$\tau_d = \frac{T_{t2}}{T_{t1}} \qquad \pi_d = \frac{P_{t2}}{P_{t1}}$$

DIFFUSER

$$\tau_c = \frac{T_{t3}}{T_{t2}} \qquad \pi_c = \frac{P_{t3}}{P_{t2}}$$

COMPRESSOR

$$\tau_b = \frac{T_{t4}}{T_{t3}} \qquad \pi_b = \frac{P_{t4}}{P_{t3}}$$

COMBUSTOR

$$\tau_t = \frac{T_{t5}}{T_{t4}} \qquad \pi_t = \frac{P_{t5}}{P_{t4}}$$

TURBINE

$$\tau_r = \frac{T_{t0}}{T_0} = 1 + \left(\frac{\gamma-1}{2}\right) M_0^2$$

$$\pi_r = \frac{P_{t0}}{P_0} = \left(1 + \left(\frac{\gamma-1}{2}\right) M_0^2\right)^{\frac{\gamma}{\gamma-1}}$$

$$\boxed{\tau_\lambda = \frac{T_{t4}}{T_0}}$$

* almost all engine performance metrics scale with this.

$$\boxed{\tau_f = h_f / c_p T_0}$$

* ratio of fuel enthalpy to ambient

Extra relations:

$$\tau_t = 1 - \frac{\tau_r(\tau_c - 1)}{(1+f)\,\tau_\lambda} = 1 - \frac{(\tau_c - 1)}{\tau_\lambda} \; ; \qquad f = \frac{\dot{m}_f}{\dot{m}_a} = \frac{\tau_\lambda - \tau_r \tau_c}{\tau_f - \tau_\lambda}$$

AA 283: Running Class Notes

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DATE 04/2026

Quick Reference Values and Schematic

$$
\begin{aligned}
C_{p\ air} &= 1005.9\ m^2/s^2K \\
R_{air} &= 287.055\ m^2/s^2K \\
P_{SL} &= 101325\ N/m^2 \\
T_{SL} &= 288.15\ K \\
\gamma = \frac{C_p}{C_v} &= 1.4 \\
R_u &= 8.3144598\ J/(mole-K) \quad \text{or} \quad kgm^2/s^2/(mole-K) \\
H = \frac{R_{air}T_{SL}}{g} &= 8434.5\ m \quad \text{scale height used for altitude determ.} \\
h_{air} &= 2.896e5\ J/kg \\
h_f &= 4.28e7\ J/kg
\end{aligned}
$$

Some tips

1. Isentropic = Adiabatic + Reversible (total values stay the same, use isentropic relations)
2. Adiabatic $\rightarrow Q = 0 \rightarrow h_t =$ conserved
3. Calorically Perfect $\rightarrow h = C_pT \quad ; \quad h_t = C_pT_t \quad ; \quad \frac{T_1}{T_2} = \frac{h_1}{h_2}$
4. Fully expanded flow $\rightarrow P_e = P_0$. **THESE ARE STATIC NOT STAGNATION VALUES**
5. Assume if there is an exit nozzle, $M_e = 1$
6. Assume $f << 1 \rightarrow \dot{m}_f << \dot{m}_a$

Chapter 1: Aircraft Drag and Propulsion Thermodynamics

0.1 Lifting Line theory basics

Various Drag Coefficients and their relationships. C_{D_i} C_{D_P} induced drag an profile drag respectively.
ϵ is span efficiency factor typically (0.7, 0.9)

$$C_D = \frac{D}{\frac{1}{2}\rho_0 U_0^2 S} \quad C_{D_P} = \frac{D_P}{\frac{1}{2}\rho_0 U_0^2 S} \quad C_{D_i} = \frac{D_i}{\frac{1}{2}\rho_0 U_0^2 S} \quad C_L = \frac{L}{\frac{1}{2}\rho_0 U_0^2 S}$$

$$C_D = C_{D_P} + C_{D_i} = C_{D_P} + \frac{C_L^2}{\pi\epsilon}\frac{S}{b^2} \approx \frac{C_{D_P}}{(1-M_0^2)^{0.5}} + \frac{C_L^2}{\pi\epsilon}\frac{S}{b^2}$$

last leg of equation above is the Prandtl-Glauert correction factor that only applies when $M_0 < 1$ accounting for compressibility factors

Here's another way to write free-stream dynamic pressure (derived from $U_0 = M_0\sqrt{\gamma RT}$ and $P = \rho RT$ (which is ideal gas equation).

$$\frac{1}{2}\rho_0 U_0^2 S = \frac{\gamma}{2} P_0 M_0^2 S$$

Aircraft sigma parameter (intrinsic)

$$\sigma = \left(\frac{S}{\pi\epsilon b^2 C_{D_P}}\right)^{\frac{1}{2}} \left(\frac{L}{\frac{\gamma}{2} P_{SL} S}\right)$$

Where L should just be weight of aircraft. Sanity sigma values usually $\sigma < 1$

From σ, we get important **non-dimensional** values

$$P = \frac{P_0}{\sigma P_{SL}} \quad \Phi = \frac{C_{D_{SL}}}{\sigma C_{D_P}} = \frac{PM_0^2}{(1-M_0^2)^{0.5}} + \frac{1}{PM_0^2}$$

Important to understand that

1. Φ is dimensionless drag and is independent of the parameters of the aircraft. It still follows that lower/higher Φ implies lower/higher true D
2. P is dimensionless pressure, but is actually a better encoder of ALTITUDE. **The HIGHER P, the LOWER altitude** h. Sanity usually $P < 10$

0.2 Maximum Mach and Altitude to Cruise

In previous section, solving $\Phi(P, M_0)$ for $P(\Phi, M_0)$ yields a quadratic that (for a given Φ) yields the following plot across many Φs

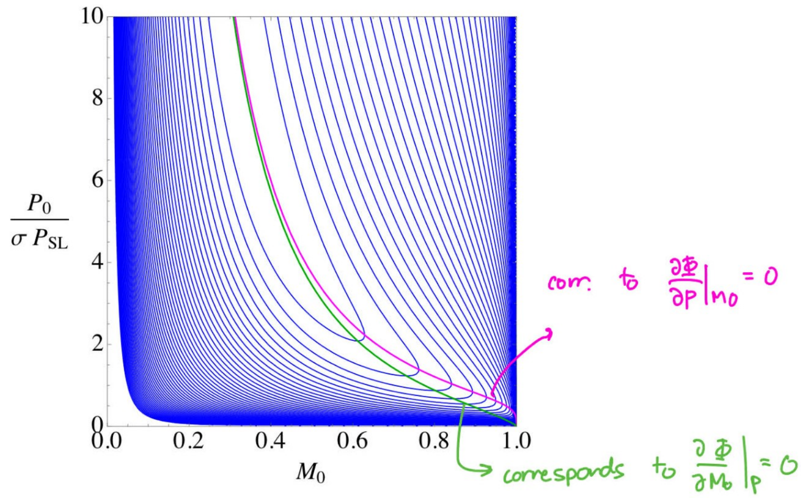


Figure 1.10: *Contours of constant Φ as a function of P and M_0 showing two paths to the cruise condition: magenta curve, maximum Mach number path, (1.25), green curve, maximum altitude path (1.29).*

The pink line corresponds to maximum Mach number at given cruise. It is given by the following equation.

$$\frac{\partial \Phi}{\partial P}\Big|_{M_0} = 0 \Rightarrow P = \frac{(1-M_0^2)^{1/4}}{M_0^2}$$

At these points of maximum Mach number correspond to the following equations.

$$\Phi = \frac{2}{(1-M_0^2)^{1/4}} \quad M_0 = \left(1 - \left(\frac{2}{\Phi}\right)^4\right)^{1/2} \quad P = \frac{\frac{2}{\Phi}}{1-(\frac{2}{\Phi})^4}$$

And a critical lift-to-drag ratio

$$L/D = \frac{1}{2}\left(\frac{\pi \epsilon b^2}{C_{D_P} S}\right)^{1/2} (1-M_0^2)^{1/4}$$

The green line corr. to maximum altitude at given cruise (fittingly corresponding to minimum P). It is given by the following equation.

$$\frac{\partial \Phi}{\partial M_0}\Big|_{P} = 0 \Rightarrow P = \frac{(1-M_0^2)^{3/4}}{M_0^2(1-\frac{1}{2}M_0^2)^{1/2}}$$

0.3 The Standard Atmosphere

The following is the textbook model for temperature ratio as a function of altitude h. **They are given for trophosphere ($h < 11,000$ m), stratosphere ($11,000 < h < 25,000$ m), and upper stratosphere**

$(h > 25,000)$. For quick reference here, $H = 8434.5\ m$

$$\frac{T_0}{T_{SL}} = (1 - 0.189961\frac{h}{H})$$
$$\frac{T_0}{T_{SL}} = 0.752 \quad \text{totally independent of } h$$
$$\frac{T_0}{T_{SL}} = (0.49259 + 0.0875168\frac{h}{H})$$

The same is provided for Pressures

$$\frac{P_0}{P_{SL}} = (1 - 0.1902742\frac{h}{H})^{5.25588}$$
$$\frac{P_0}{P_{SL}} = 0.22332\ \exp(1.727 - 1.3242\frac{h}{H})$$
$$\frac{P_0}{P_{SL}} = 0.02479\left(\frac{T_0}{216.69}\right)^{-11.388}$$

Note that it is much more effective to map pressures to altitudes because it is actually 1-1, unlike temperature. This is evident from the plots below.

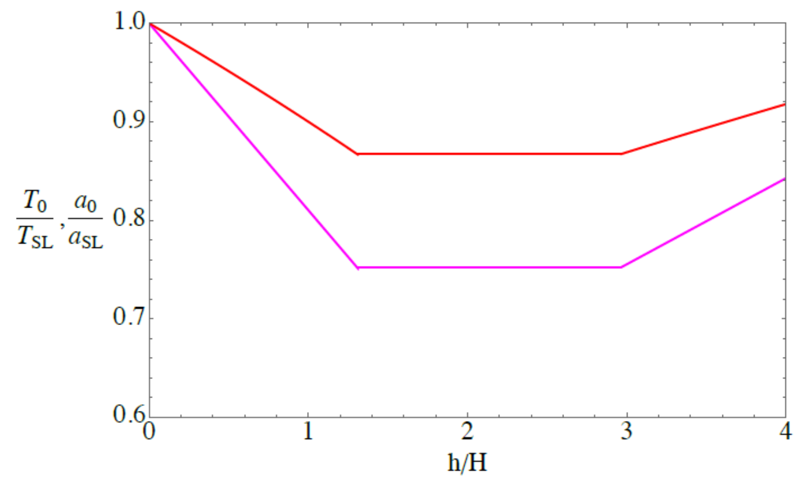


Figure 1.12: *Standard Atmosphere temperature distribution (magenta) and speed of sound (red) through the beginning of the upper stratosphere.*

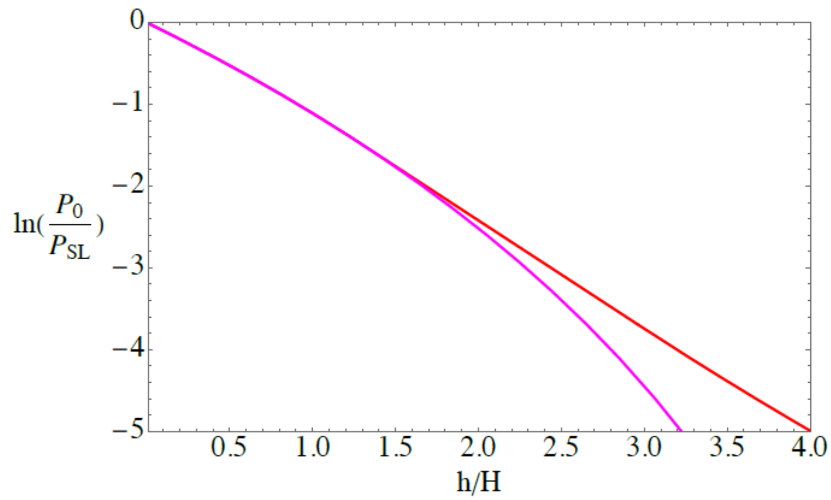


Figure 1.14: *Standard atmospheric pressure versus height through the beginning of the upper stratosphere. The magenta curve is Equation* (1.36).

As a just-in-case, here's some stuff for visocity and thermal conductivity at different h

Viscosity and Conductivity

The viscosity and thermal conductivity of an ideal gas depends excusively on temperature. Sutherland's formula for viscosity is

$$\frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \left(\frac{T_{ref} + T_{s\mu}}{T + T_{s\mu}}\right) \tag{1.40}$$

where $\mu_{ref} = 1.716 \times 10^{-5}\, N sec/m^2$, $T_{ref} = 273.15\, K$ and $T_{s\mu} = 110.4\, K$.

and for the thermal conductivity

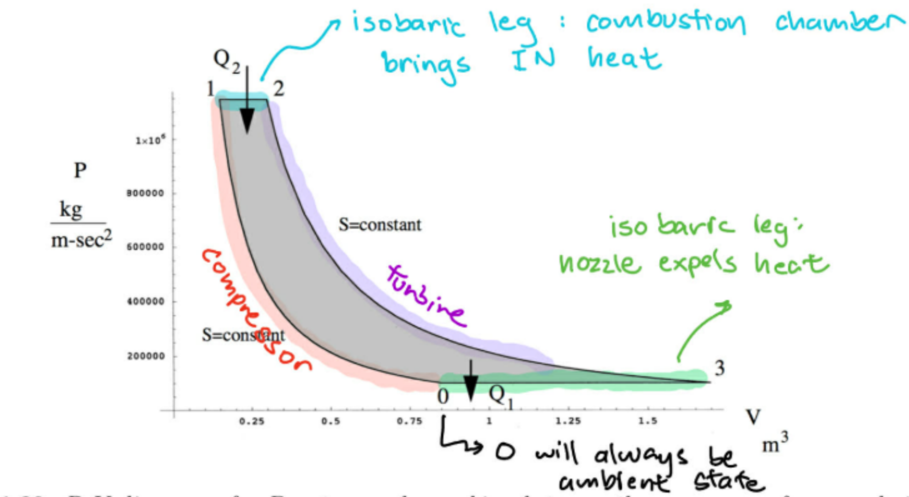
$$\frac{k(T)}{k_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \left(\frac{T_{ref} + T_{sk}}{T + T_{sk}}\right) \tag{1.41}$$

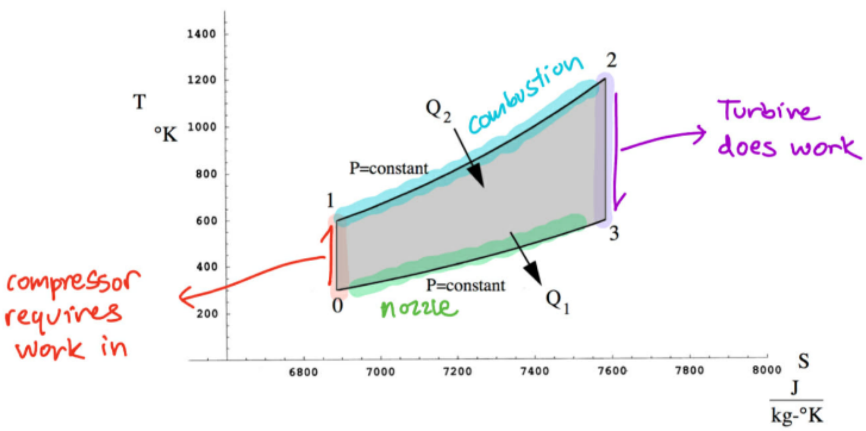
where $k_{ref} = 2.41 \times 10^{-2}\, J/(m\, K\, sec)$, $T_{ref} = 273.15\, K$ and $T_{sk} = 194.0\, K$.

Sutherland model

0.4 Brayton Cycle

For the sake of this course, most engines are modeled as a Brayton Cycle.





The work done by a Brayton cycle is calculated via the following

$$W = Q_2 - Q_1 = (H_2 - H_1) - (H_3 - H_0)$$

The efficiency of a Brayton cycle is

$$\eta_B = 1 - \frac{Q_1}{Q_2} = 1 - \frac{H_3 - H_0}{H_2 - H_1} = 1 - \frac{T_0}{T_1}$$

NOTE THAT THESE NUMBER SUBSCRIPTS ARE NOT ASSOCIATED WITH ENGINE STATIONS

also note that this is different from the definition of thermodynamic efficiency η_{th} from chapter 2

Chapter 2: Engine Performance Parameters

0.5 Thrust and energy balance

First, establish that the exit mass flow is given by

$$\rho_e U_e A_e = \dot{m}_a + \dot{m}_f$$

Skipping much of the derivations, the thrust equation for ALL general propulsion systems boils down to the following.

$$T = \dot{m}_a(U_e - U_0) + (P_e - P_0)A_e + \dot{m}_f U_e$$

Notice that the three terms here are essentially

1. momentum change of air mass
2. exit pressure differential
3. momentum change of fuel mass

Energy balance across an engine (assuming all processes adiabatic). For quick reference, $h_{t0} = h_{air} = 2.896e5\ J/kg$ and $h_f = 4.28e7\ J/kg$. Note $\frac{h_f}{h_{air}} = 148!$

$$(\dot{m}_a + \dot{m}_f)h_{te} = \dot{m}_a h_{t0} + \dot{m}_f h_f$$

The equation above is very useful across stations 3 and 4 of a burner.

0.6 Efficiencies

We cover many different definitions of efficiency

1. Overall Efficiency η_{ov}: self-explanatory
2. Propulsive Efficiency η_{pr}: an aerodynamic factor to overall efficiency
3. Thermal Efficiency η_{th}: a thermodynamic factor to overall efficiency
4. Polytropic Efficiency η_{pc}: Efficiency of an infinitesimal compression (instantaneous efficiency)
5. Compressor Efficiency η_c: Efficiency of compressor
6. Turbine Efficiency η_e: Efficiency of turbine

Definition of overall efficiency (both mathematical and conceptual)

$$\eta_{ov} = \frac{TU_0}{\dot{m}_f h_f} = \frac{\text{power to vehicle}}{\text{tot. energy released per sec of combustion}}$$

A potentially easier way to determine η_{ov} is $\eta_{ov} = \eta_{pr} * \eta_{th}$
Definition of Propulsive efficiency

$$\eta_{pr} = \frac{\text{Power to vehicle}}{\text{Power to vehicle} + \Delta\text{KE of air per sec} + \Delta\text{KE of fuel per sec}}$$

Which, when exhaust is fully expanded and $f << 1$, boils down to

$$\eta_{pr} = \frac{2U_0}{U_e + U_0}$$

This equation is insightful, it implies that η_{pr} is maxed when $U_e >> U_0$, meaning it is more efficient to move a LOT of air at a SMALL ΔU than the other way around.
Definition of Thermodynamic efficiency

$$\eta_{th} = \frac{\text{Power to vehicle} + \Delta\text{KE of air per sec} + \Delta\text{KE of fuel per sec}}{\dot{m}_f h_f}$$

Again assuming full nozzle expansion this simplifies to a form where $\eta_{th}(Ue, U_0)$.

$$\eta_{th} = \frac{(\dot{m}_a + \dot{m}_f)\frac{U_e^2}{2} - \dot{m}_a\frac{U_0^2}{2}}{\dot{m}_f h_f}$$

We can also massage to obtain $\eta_{th}(h_e, h_0)$ by using the relation $h_{te} = h_e + \frac{U_e^2}{2}$ and $h_{t0} = h_0 + \frac{U_0^2}{2}$. This yields

$$\eta_{th} = 1 - \frac{Q_{rejected}}{Q_{input}} = 1 - \frac{(\dot{m}_a + \dot{m}_f)(h_e - h_0) + \dot{m}_f h_0}{\dot{m}_f h_f}$$

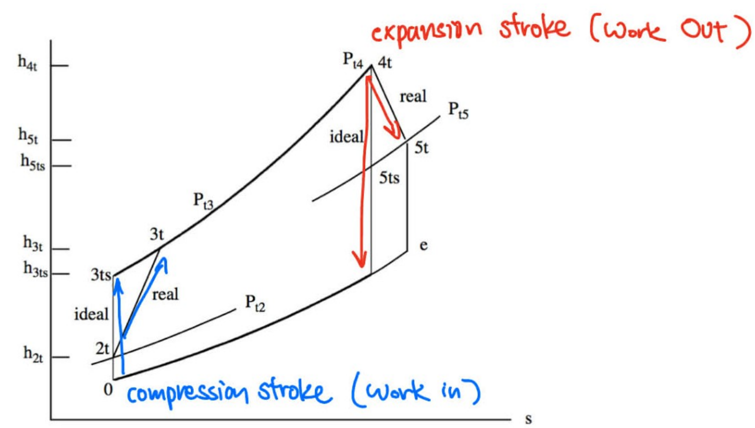


Figure 2.5: *Path of a turbojet in h-s coordinates with non-ideal compressor and turbine.*

Looking at the above figure, we have the following definitions for η_c and η_e, the last equation of which **assumes a calorically perfect gas**.

$$\eta_c = \frac{\text{Work in needed to reach } P_{t3}/P_{t2} \text{ isentropically}}{\text{Actual work in needed to reach } P_{t3}/P_{t2}} = \frac{h_{t3s} - h_{t2}}{h_{t3} - h_{t2}} = \frac{T_{t3s} - T_{t2}}{T_{t3} - T_{t2}}$$

$$\eta_e = \frac{\text{Actual work out needed to reach } P_{t5}/P_{t4}}{\text{Work out needed to reach } P_{t5}/P_{t4} \text{ isentropically}} = \frac{h_{t5} - h_{t4}}{h_{t5s} - h_{t4}} = \frac{T_{t5} - T_{t4}}{T_{t5s} - T_{t4}}$$

Finally, we realize that as-defined, η_c and η_e do not apply well across different compressors with different compressor ratios. Instead, the polytropic efficiency η_{pc} and η_{pe} exist for this case. These efficiencies are defined and the original efficiencies η_c and η_e are expressed as a function of the polytropic efficiencies below.

$$\eta_{pc} = \frac{\gamma - 1}{\gamma} \frac{\ln \frac{P_{t3}}{P_{t2}}}{\ln \frac{T_{t3}}{T_{t2}}} \qquad \eta_{ec} = \frac{\gamma}{\gamma - 1} \frac{\ln \frac{T_{t5}}{T_{t4}}}{\ln \frac{P_{t5}}{P_{t4}}}$$

$$\eta_c = \frac{\frac{P_{t3}}{P_{t2}}^{\frac{\gamma-1}{\gamma}} - 1}{\frac{P_{t3}}{P_{t2}}^{\frac{\gamma-1}{\gamma\eta_{pc}}} - 1} \qquad \eta_e = \frac{\frac{P_{t5}}{P_{t4}}^{\frac{\eta_{pe}(\gamma-1)}{\gamma}} - 1}{\frac{P_{t5}}{P_{t4}}^{\frac{\gamma-1}{\gamma}} - 1}$$

0.7 Bregeut Aircraft Range

This is an aside that is used as a quick calc for how far an aircraft can travel. Assume cruise conditions where L/D and η_{ov} are constant.

$$R = \eta_{ov} \frac{h_f}{g} \frac{L}{D} \ln(\frac{w_{init}}{w_{fin}})$$

Intuitively, we can determine aircraft cruising range if we have efficiency, L/D ratio, energy storage density of fuel, and amount of fuel used (w_{init} and w_{fin})

0.8 Specific Impulse, Fuel Consumption, and Important Dim-less values

The following are going to be textbook formulas without much explanation.

$$I_{sp} = \frac{\text{Thrust}}{\text{Fuel flow}} = \frac{T}{\dot{m}_f g} \quad \text{unitless specific impulse}$$

$$SFC = \frac{\text{lbs fuel burned per hour}}{\text{lbs Thrust}} = \frac{3600}{I_{sp}} \quad \text{unitless specific fuel consumption.}$$

$$\frac{T}{P_0 A_0} = \gamma M_0^2 \left((1+f)\frac{U_e}{U_0} - 1 \right) + \frac{A_e}{A_0} \left(\frac{P_e}{P_0} - 1 \right) \quad \text{Pressure-normalized dimensionless thrust}$$

$$\frac{T}{\dot{m}_a a_0} = \frac{1}{\gamma M_0} \frac{T}{P_0 A_0} \quad \text{Momentum-normalized dimensionless thrust.}$$

$$\frac{I_{sp} g}{a_0} = \frac{1}{f} \frac{T}{\dot{m}_a a_0} \quad \text{Dimensionless specific impulse}$$

Lastly, you can also express overall efficiency as a function of dimensionless thrust

$$\eta_{ov} = \frac{\gamma - 1}{\gamma} \frac{1}{f \tau_f} \frac{T}{P_0 A_0}$$

Where $\tau_f = \frac{h_f}{C_p T_0}$ is the ratio of enthalpy of the fuel to the ambient environment.

0.9 Engine Notation

This section has its own hand-written cheatsheet due to being schematic heavy

Chapter 3: The Ramjet Cycle

When analyzing a Ramjet, it is very important to keep Rayleigh Flow and Rankine Hugoniot conditions in mind. Also, it is useful to keep this simplified expression for mass conservation applied all throughout the channel

$$\frac{P_{t0}A_0}{\sqrt{T_{t0}}}f(M_0) = \frac{P_{te}A_e}{\sqrt{T_{te}}}f(M_e)$$

Notice that according to mass conservation, if you add heat and T_t goes up, P_t must go down.

0.10 Ideal Ramjet

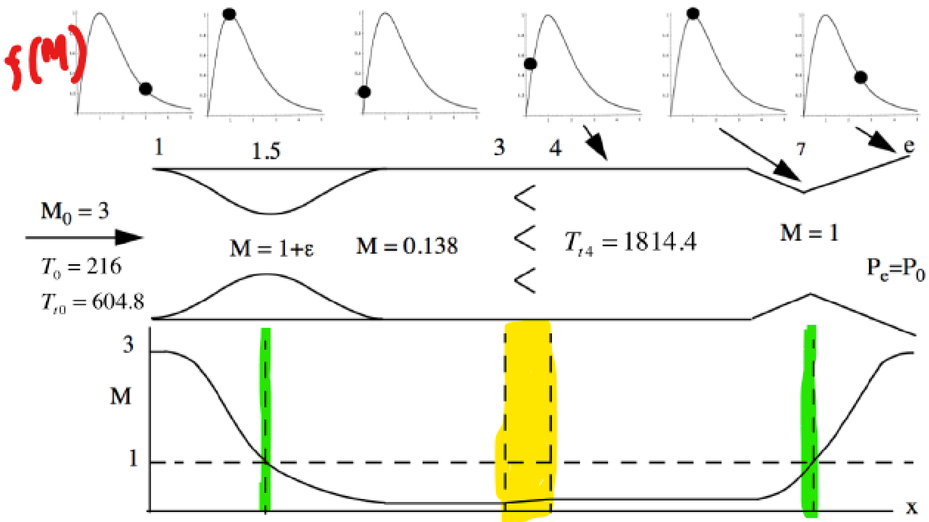


Figure 3.11: *Ideal ramjet with a fully expanded nozzle.*

Much of the chapter up until this point is building up the design for an ideal ramjet. Here are a few key points.

1. The ideal ramjet comes with the implications that $P_{te} = P_{t0}$, $M_e = M_0$, and $\frac{T_{te}}{T_{t0}} = \frac{T_e}{T_0}$.
2. from conservation of total pressure, we can see that the ideal ramjet is designed for minimal pressure losses.
3. The converging part of the inlet exists to bring the incoming flow subsonic, where its heat capacity increases greatly and it is primed for combustion.
4. **the engine ideally has a VERY WEAK normal shock at the throat of the inlet. The ramjet is considered unstarted when this shock moves further upstream to the converging section.**
5. The diverging part of the inlet exists to further slow the now-subsonic flow and create a surface for propulsive forces to react upon.
6. The amount of heat addition from fuel is minimized

7. The converging part of the exit nozzle exists to push the 1st shock to the inlet throat and also create a second weak shock at the exit nozzle throat.

8. The diverging part of the exit nozzle exists to fully-expand the flow ($P_e = P_0$).

9. For the purposes of this class, all ramjets that are not unstarted will NOT have the concept of upstream capture area.

10. Ramjets practically operate best at Mach 3-6.

0.11 Optimization of an Ideal Ramjet

For ramjet optimization, it is easiest to optimize dimensionless thrust based on cycle parameters τ_r and τ_f via the following equation.

$$\frac{T}{P_0 A_0} = \frac{2\gamma}{\gamma - 1}(\tau_r - 1)\left(\frac{\tau_f - \tau_r}{\tau_f - \tau_\lambda}\sqrt{\frac{\tau_\lambda}{\tau_r}} - 1\right)$$

Take the partial derivative of the equation above set to zero and you get the fuel burning ratio τ_λ as a function of optimal ambient condition $\tau_{r \text{ max thrust}}$.

$$\frac{\partial}{\partial \tau_r}\frac{T}{P_0 A_0} = 0 \quad \Rightarrow \quad \tau_\lambda^{1/2} = \frac{2(\tau_{r \text{ max thrust}})^{3/2}}{\tau_{r \text{ max thrust}} + 1}$$

Out of formality, the dimensionless specific impulse is also given as a function of engine parameters.

$$\frac{I_{sp} g}{a_0} = \frac{\left(\frac{2\gamma}{\gamma-1}(\tau_r - 1)\right)^{1/2}}{\frac{\tau_\lambda - \tau_r}{\tau_f - \tau_\lambda}}\left(\frac{\tau_f - \tau_r}{\tau_f - \tau_\lambda}\sqrt{\frac{\tau_\lambda}{\tau_r}} - 1\right)$$

0.12 Ramjet Control

Only 2 main modes of ramjet control, both of them are expressed below in the direction that HELPS in generating dimensionless thrust.

1. **Decrease nozzle exit area** A_e: causes P_{te} to rise and the inlet shock moves towards the throat and gets weaker and weaker.

2. **Increae fuel burning temperature** $T_{te} = T_{t4}$: has the same effect

0.13 Scramjets

The main problem with a scramjet is that the the coefficient of thrust tends to f as Mach number tends to infinity. This is a problem because f can be incredibly small.

$$C_{thrust} = \frac{T}{\frac{1}{2}\rho_0 U_0^2 A_0} = 2\left(\left(\frac{f(1+f)\tau_f}{1 + \frac{\gamma-1}{2}M_0^2} + 1 + f\right)^{1/2} - 1\right)$$

And

$$\lim_{M_0 \to \infty} C_{thrust} = f$$

Chapter 4: The Turbojet Cycle

0.14 Thermodynamic efficiency of a turbojet

For turbojets, where typically compressor and turbine share the same lossless shaft, we have the following conservation equation.

$$(\dot{m}_a + \dot{m}_f)(h_{t4} - h_{t5}) = \dot{m}_a(h_{t3} - h_{t2})$$

Across the engine, the enthalpy balance hence becomes

$$(\dot{m}_a + \dot{m}_f)h_{t5} = \dot{m}_a h_{t2} + \dot{m}_f h_f$$

and is equivalent to the following assuming adiabatic nozzle and inlet

$$(\dot{m}_a + \dot{m}_f)h_{te} = \dot{m}_a h_{t0} + \dot{m}_f h_f$$

The thermal efficiencies specific to this case are screenshotted and included below.

Now the thermal efficiency (4.3) can be written as

$$\eta_{th} = \frac{(\dot{m}_a + \dot{m}_f)(h_{te} - h_e) - \dot{m}_a(h_{t0} - h_0)}{(\dot{m}_a + \dot{m}_f)h_{t4} - \dot{m}_a h_{t3}}. \tag{4.10}$$

Using (4.9) and (4.7), equation (4.10) becomes

$$\eta_{th} = \frac{(\dot{m}_a + \dot{m}_f)h_{t4} - \dot{m}_a h_{t3} - (\dot{m}_a + \dot{m}_f)h_e + \dot{m}_a h_0}{(\dot{m}_a + \dot{m}_f)h_{t4} - \dot{m}_a h_{t3}} \tag{4.11}$$

or

$$\begin{aligned}
\eta_{th} &= 1 - \frac{Q_{rejected\,during\,the\,cycle}}{Q_{input\,during\,the\,cycle}} = 1 - \left(\frac{(\dot{m}_a + \dot{m}_f)h_e - \dot{m}_a h_0}{(\dot{m}_a + \dot{m}_f)h_{t4} - \dot{m}_a h_{t3}}\right) \\
\eta_{th} &= 1 - \frac{h_0}{h_{t3}}\left(\frac{(1+f)\frac{h_e}{h_0} - 1}{(1+f)\frac{h_{t4}}{h_{t3}} - 1}\right).
\end{aligned} \tag{4.12}$$

0.15 Ideal Turbojet

This is how pressure evolves through a turbojet using engine parameters

$$P_{te} = P_0 \pi_r \pi_d \pi_c \pi_b \pi_t \pi_n$$

In an IDEAL turbojet, **the diffuser, nozzle, AND combustion chamber are all assumed lossless**, AKA.

$$\pi_d = \pi_b = \pi_n = 1$$

Finally, applying the assumptions of **isentropy and full nozzle expansion**, we are left with

$$\pi_r \pi_c \pi_t = \left(1 + \frac{\gamma - 1}{2} M_e^2\right)^{\frac{\gamma}{\gamma - 1}}$$

$$\pi_c = \tau_c^{\frac{\gamma}{\gamma-1}} \quad ; \quad \pi_t = \tau_t^{\frac{\gamma}{\gamma-1}}$$

The Mach number ratio is thus

$$\frac{M_e^2}{M_0^2} = \frac{\tau_r\tau_c\tau_t - 1}{\tau_r - 1}$$

Applying similar principles to Temperature, we have

$$T_{te} = T_0\tau_r\tau_d\tau_c\tau_b\tau_t\tau_n = T_0\tau_r\tau_c\tau_b\tau_t = T_e\tau_r\tau_c\tau_t \quad \Rightarrow \quad \frac{T_e}{T_0} = \tau_b = \frac{T_{t4}}{T_{t3}} = \frac{\tau_\lambda}{\tau_r\tau_c}$$

Some additional formulas for the ideal turbojet are given below

Our thrust formula is now

$$\frac{T}{P_0A_0} = \frac{2\gamma}{\gamma - 1}(\tau_r - 1)\left((1+f)\left(\left(\frac{\tau_r\tau_c\tau_t - 1}{\tau_r - 1}\right)\frac{\tau_\lambda}{\tau_r\tau_c}\right)^{1/2} - 1\right). \tag{4.33}$$

The fuel/air ratio is found from

$$f = \frac{\tau_\lambda - \tau_r\tau_c}{\tau_f - \tau_\lambda}. \tag{4.34}$$

turbine. They are related by the work matching condition (4.6) repeated here in terms of the temperatures.

$$(\dot{m}_a + \dot{m}_f)\, C_P\, (T_{t4} - T_{t5}) = \dot{m}_a C_P\, (T_{t3} - T_{t2}) \tag{4.36}$$

For simplicity we have assumed the same value of C_p for the compressor and turbine. If we divide (4.36) by $C_p T_0$ then it becomes

$$(1 + f)\, \tau_\lambda\, (1 - \tau_t) = \tau_r\, (\tau_c - 1) \tag{4.37}$$

or

$$\tau_t = 1 - \frac{\tau_r\, (\tau_c - 1)}{(1 + f)\, \tau_\lambda} \tag{4.38}$$

where we have assumed $T_{t2} = T_{t0}$. The result (4.38) only assumes adiabatic flow in the inlet and no shaft losses. It is not tied to the other assumptions of the ideal cycle. The velocity ratio across the engine is now

$$\left(\frac{U_e}{U_0}\right)^2 = \frac{1}{(\tau_r - 1)} \left(\tau_\lambda - \tau_r\, (\tau_c - 1) - \frac{\tau_\lambda}{\tau_r \tau_c}\right) \tag{4.39}$$

where the fuel/air ratio has been neglected.

0.16 Engine-Matching conditions

In general, these are the equations that arise out of mass-flow matching across different turbojet engine components.

$$\begin{aligned}
\tau_t &= \left(\frac{A_4^*}{A_8}\right)^{\frac{2(\gamma-1)}{\gamma+1}} \\
\tau_c - 1 &= \frac{\tau_\lambda}{\tau_r}(1 - \tau_t) \\
f(M_2) &= \frac{\pi_c}{\sqrt{\tau_\lambda/\tau_r}} \frac{A_4^*}{A_2} \\
f(M_2) &= \frac{1}{\pi_d} \frac{A_0}{A_2} f(M_0)
\end{aligned}$$

Qualitatively, these relations are respectively derived from

1. turbine-nozzle mass flow match, hence it depends on stations 4 and 8/e
2. not a mass-balance term, but **it is important to remember that $\tau_{t,r,c,\lambda}$ are related like this (see eq. 4.38)**
3. compressor-turbine mass balance
4. free-stream to compressor mass balance

Chapter 5: The Turbofan Cycle

A Turbofan is essentially a turbojet with a bypass component that does not go through combustion and is simply accelerated by a fan.

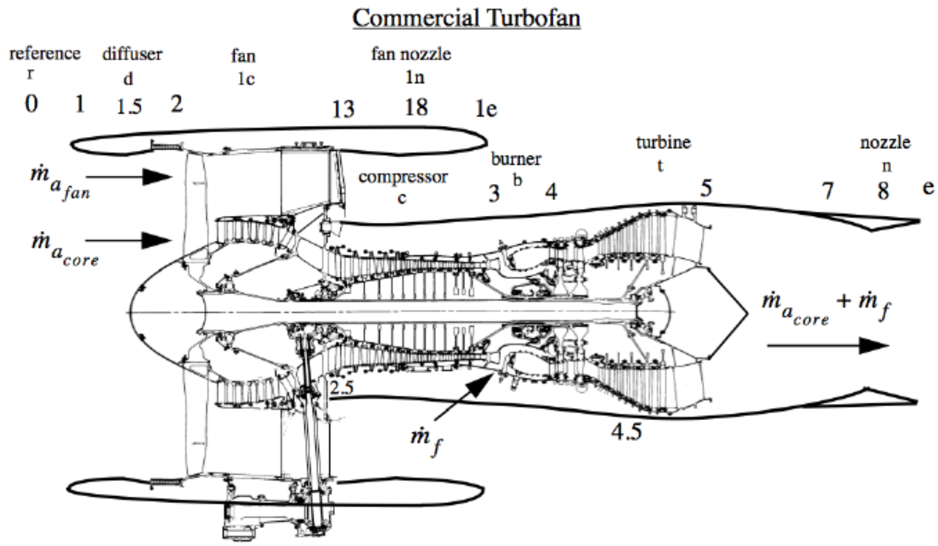


Figure 1: Schematic of a classic Turbofan

The thrust equation for a turbofan is only slightly modified from the general thrust equation with bypass values included.

$$T = \dot{m}_{a_{core}}(U_e - U_0) + \dot{m}_{a_{fan}}(U_{e1} - U_0) + \dot{m}_f U_e + (P_e - P_0)A_e + (P_{e1} - P_0)A_{1e}$$

The air and fuel mass equations are as such

$$\dot{m}_a = \dot{m}_{a_{core}} + \dot{m}_{a_{fan}} \quad ; \quad f = \frac{\dot{m}_f}{\dot{m}_a}$$

The **bypass fraction** B and **bypass ratio** β are important to the turbofan.

$$B = \frac{\dot{m}_{a_{fan}}}{\dot{m}_{a_{fan}} + \dot{m}_{a_{core}}} \quad ; \quad \beta = \frac{\dot{m}_{a_{fan}}}{\dot{m}_{a_{core}}} \quad ; \quad \beta = \frac{B}{1-B} \quad ; \quad B = \frac{\beta}{1+\beta}$$

The Ideal Turbofan

All the assumptions for an ideal turbojet hold, an additional isentropic assumptions are added for the fan duct.

$$\pi_d, \pi_b, \pi_n, \pi_{n1} = 1 \quad ; \quad P_e = P_0 \quad ; \quad P_{e1} = P_0 \quad ; \quad \pi_{c,t,c1} = \tau_{c,t,c1}^{\frac{\gamma}{\gamma-1}}$$

The dimensionless thrust in terms of bypass ratio (assuming $f \ll 1$) is

$$\frac{T}{\dot{m}_a a_0} = M_0 \left(\left(\frac{1}{1+\beta} \right) \left(\frac{U_e}{U_0} - 1 \right) + \left(\frac{\beta}{1+\beta} \right) \left(\frac{U_{e1}}{U_0} - 1 \right) \right)$$

Ideal Fan Stream

$$
\begin{aligned}
P_{te1} &= P_0 \pi_r \pi_{c1} = P_{e1}\left(1+\frac{\gamma-1}{2}M_{e1}^2\right)^{\frac{\gamma}{\gamma-1}}\\
T_{te1} &= T_0 \tau_r \tau_{c1} = T_0\left(1+\frac{\gamma-1}{2}M_{e1}^2\right) = T_{e1}\left(1+\frac{\gamma-1}{2}M_{e1}^2\right)\\
\left(\frac{U_{e1}}{U_0}\right)^2 &= \left(\frac{M_{e1}}{M_0}\right)^2 = \frac{\tau_r\tau_{c1}-1}{\tau_r-1}
\end{aligned}
$$

Ideal Core Stream

$$
\begin{aligned}
P_{te} &= P_0 \pi_r \pi_c \pi_t = P_e\left(1+\frac{\gamma-1}{2}M_e^2\right)^{\frac{\gamma}{\gamma-1}}\\
T_{te} &= T_0 \tau_r \tau_c \tau_b \tau_t = T_e\left(1+\frac{\gamma-1}{2}M_e^2\right) = T_e\tau_r\tau_c\tau_t\\
\frac{T_e}{T_0} &= \tau_b = \frac{\tau_\lambda}{\tau_r\tau_c}\\
\left(\frac{U_{e1}}{U_0}\right)^2 &= \frac{\tau_r\tau_c\tau_t-1}{\tau_r-1}\frac{\tau_\lambda}{\tau_r\tau_c} = \left(\frac{M_{e1}}{M_0}\right)^2\frac{\tau_\lambda}{\tau_r\tau_c}
\end{aligned}
$$

Turbine-Compressor-Fan matching

In a turbofan, the work generated by the turbine is consumed by both the compressor AND the fan. hence we have an extended version of the work-matching equation.

$$
\left(\dot{m}_{a_{\text{core}}}+\dot{m}_f\right)\left(h_{t4}-h_{t5}\right) = \dot{m}_{a_{\text{core}}}\left(h_{t3}-h_{t2}\right) + \dot{m}_{a_{\text{fan}}}\left(h_{t31}-h_{t2}\right)
$$

For an approximation of $f \ll 1$, we have the following matching condition relation.

$$
\tau_t = 1 - \frac{\tau_r}{\tau_\lambda}\left(\left(\tau_c-1\right)+\beta\left(\tau_{c1}-1\right)\right)
$$

The fuel-air ratio is as follows, derived from energy balance across burner.

$$
f = \left(\frac{1}{1+\beta}\right)\frac{\tau_\lambda-\tau_r\tau_c}{\tau_f-\tau_\lambda}.
$$

Specific Impulse of a Turbofan

$$
\frac{I_{sp}g}{a_0} = M_0\left(\frac{\tau_f-\tau_\lambda}{\tau_\lambda-\tau_r\tau_c}\right)\left(\left(\frac{U_e}{U_0}-1\right)+\beta\left(\frac{U_{e1}}{U_0}-1\right)\right).
$$

Chapter 6: The Turboprop Cycle

Turboprops are effectively considered turbofan engines that have full bypass. They importantly differ from just simply being considered p̈ropellersb̈ecause they are powered by a turbojet rather than a traditional piston engine.

Turboprops are actually the most efficient propulsion system at low Mach numbers. However they suffer the following practical downsides.

1. Propellers cannot have a tip speed that is supersonic. This creates a local shock that causes nightmares for structure and noise.
2. If propellers must remain slow relative to the turbojet turbine driving it, a HUGE reduction gearbox must be used. These gears fail constantly and are very heavy.
3. The turboprop gets extremely noisy in the transonic and supersonic flight range.

The following if the reference figure for all Turboprop analysis.

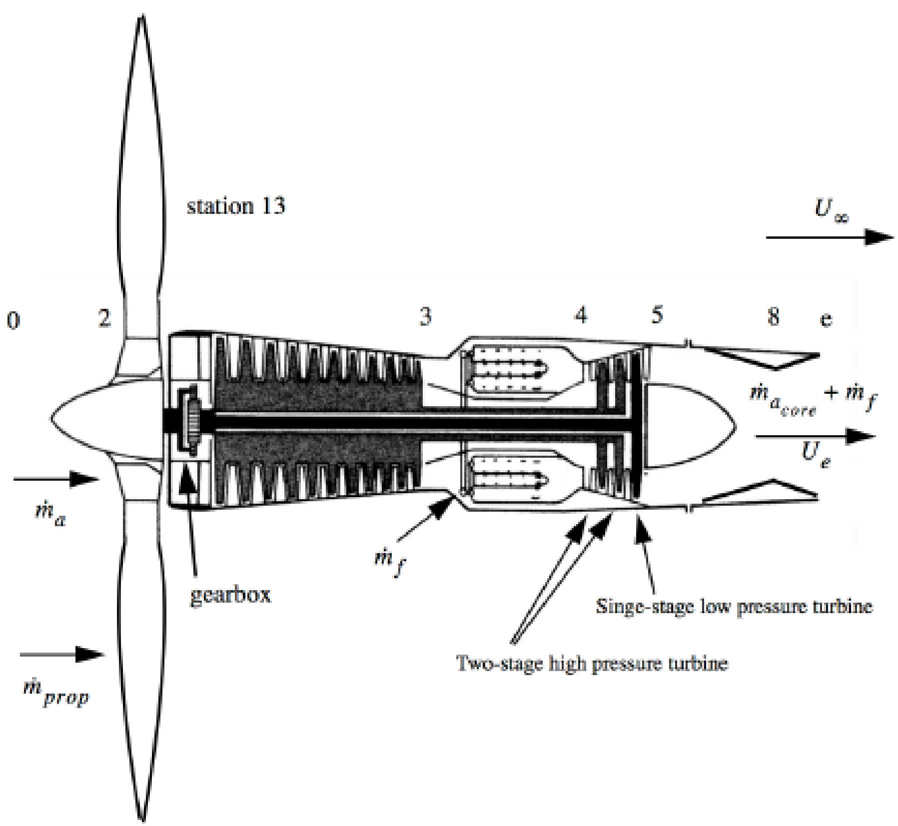


Figure 2: Schematic of a Turboprop

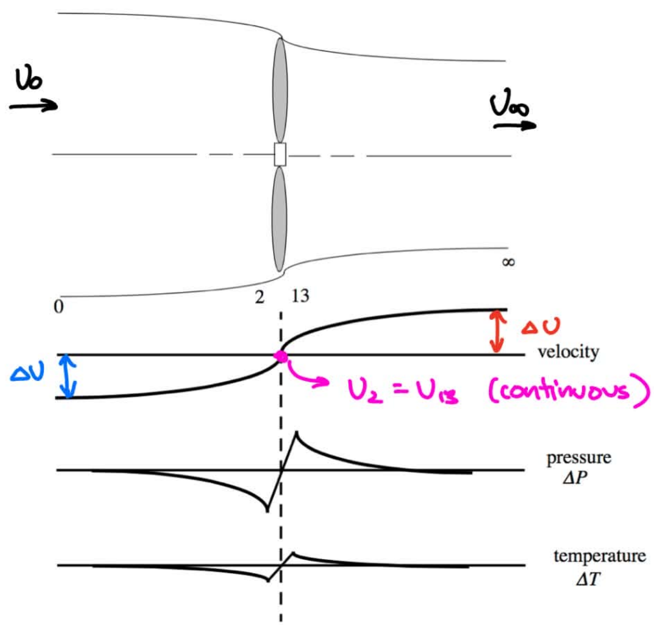
Preliminary: Froude's Theorem

Froude's theorem was proved in homework via Bernoulli's equation (applicable because we are subsonic) and mass conservation. It essentially states **the velocity change induced by a propeller is the SAME upstream and downstream.**

Froude's Thm:

$$U_2 = U_{13} = \frac{U_0 + U_\infty}{2}.$$

AKA $\Delta U = \Delta U$



Propeller Efficiency

Propeller effeciency is most roughly defined as

$$\eta_{prop} = \frac{T_{prop} U_0}{W_p}$$

Where the numerator is the output power of the propeller that actually went into propulsion, and the denominator W_p is the power supplied TO the propeller via the input shaft from the turbojet.

T_{prop} and W_p are defined as

$$T_{prop} = \dot{m}_{prop}(U_\infty - U_0)$$
$$W_p = \dot{m}_{prop}(h_{t13} - h_{t2}) = \dot{m}_{prop} C_p (T_{13} - T_2)$$

The P, P_t and T, T_t relationship pre and post-propeller are given by the following relation involving the polytropic efficiency η_{pc}

$$\frac{P_{t13}}{P_{t2}} = \left(\frac{T_{t13}}{T_{t2}}\right)^{\frac{\gamma \eta_{pc}}{\gamma - 1}} \quad ; \quad \frac{P_{13}}{P_2} = \left(\frac{T_{13}}{T_2}\right)^{\frac{\gamma \eta_{pc}}{\gamma - 1}}$$

Notice that these two relations are the same because $U_{13} = U_2$ means static value and total value ratios are the same.

Lastly, here are some potentially useful additional definitions for η_{prop} derived from the above relations.

$$\eta_{prop} = \left(\frac{2U_0}{U_0 + U_\infty}\right)\left(\frac{\frac{1}{2}\dot{m}_{prop}(U_\infty^2 - U_0^2)}{W_p}\right) = \left(\frac{2U_0}{U_0 + U_\infty}\right)\eta_{pc} = \eta_{pr}\eta_{pc}$$

Where, recall $\eta_{pr} = \left(\frac{2U_0}{U_0+U_\infty}\right)$ is the classic propulsive efficiency.

Work Coefficient C and Power Balance W_p

First, just as in a Turbofan we can describe total thrust of a turboprop as

$$T_{total} = T_{core} + T_{prop} = \dot{m}_a(U_e - U_0) + \dot{m}_f U_e + (P_e - P_0)A_e + \dot{m}_{prop}(U_\infty - U_0)$$

The work output coefficients C exist as performance metrics of a turboprop engine because of a peculiarity of efficiency definition at low airspeeds U_0. The relations are as follows.

$$\begin{aligned}
C_{total} &= C_{core} + C_{prop} = \frac{T_{total}U_0}{\dot{m}_a C_p T_0} \\
C_{core} &= (\gamma - 1)M_0^2\left((1+f)\frac{M_e}{M_0}\sqrt{\frac{T_e}{T_0}} - 1\right) \\
C_{prop} &= \eta_{prop}\frac{W_p}{\dot{m}_a C_p T_0} \\
\frac{T}{P_0 A_0} &= \frac{\gamma}{\gamma - 1}C_{total}
\end{aligned}$$

In these relations, you can think of $\dot{m}_a C_p T_0$ as the total incoming enthalpy of all the air hitting the system.

So far, we have only considered the propeller-side of W_p. Now we consider the turbojet-side of W_p, which is where the power is generated. As expected, it depends on gearbox efficiency η_g and shaft efficiency η_m.

$$\begin{aligned}
W_p &= \eta_g((\dot{m}_a + \dot{m}_f)\eta_m(h_{t4} - h_{t5}) - \dot{m}_a(h_{t3} - h_{t2})) \\
C_{prop} &= \eta_{prop}\eta_g((1+f)\eta_m\tau_\lambda(1 - \tau_t) - \tau_r(\tau_c - 1))
\end{aligned}$$

Ideal Turboprop

**see the section on the ideal turbojet, the assumptions and relations are exactly the same*

Chapter 7: Rockets

Rocket engine analysis is governed much more by fluid mechanics than thermodynamic cycles like a turbojet. Here is a basic conceptual schematic of a rocket engine (important values highlighted).

Thrust and Momentum

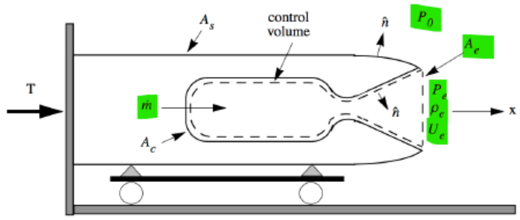


Figure 7.1: *Rocket thrust schematic.*

Other quantities defined in Figure 7.1 are as follows.

$$\begin{aligned}
&A_s = outside\ surface\ of\ the\ vehicle\ exposed\ to\ P_0 \\
&A_c = inside\ surface\ of\ the\ combustion\ chamber \\
&A_e = nozzle\ exit\ area \\
&\hat{n} = outward\ unit\ normal \\
&P_e = area\ averaged\ exit\ gas\ pressure \\
&\rho_e = area\ averaged\ exit\ gas\ density \\
&U_e = area\ averaged\ x-component\ of\ velocity\ at\ the\ nozzle\ exit
\end{aligned} \tag{7.1}$$

Figure 3: Rocket engine schematic

The thrust equation for a rocket is derived from control-volume analysis and contains many intermediate integrals. The final algebraic form is

$$\begin{aligned}
T &= \rho_e U_e^2 A_e + (P_e - P_0)A_e \\
&= \dot{m}U_e + (P_e - P_0)A_e \\
\dot{m} &= \rho_e U_e A_e
\end{aligned}$$

If it is stated that an engine nozzle is fully-expanded, that means $P_e = P_0$ and the 2nd term dies to 0.

Momentum balance is given by the following intuitive form.

$$\begin{aligned}
M_r(t)\frac{dV_r(t)}{dt} &= \rho_e U_e^2 A_e + (P_e - P_0)A_e - D \\
&= T - D
\end{aligned}$$

Effective Exhaust Velocity C and Specific Impulse I_{sp}

Recall the following two fundamental definitions for impulse I and mass of propellant expended M_p

$$I = \int_0^t T dt \quad ; \quad M_p = \int_0^t \dot{m} dt$$

With these, we can define the effective exhaust velocity C

$$
\begin{aligned}
C = \frac{dI}{dM_p} = \frac{T}{\dot{m}} &= U_e + \frac{A_e}{\dot{m}}(P_e - P_0) \\
&= U_e\left(1 + \frac{P_e A_e}{\rho_e U_e^2 A_e}\left(1 - \frac{P_0}{P_e}\right)\right) \\
&= U_e\left(1 + \frac{1}{\gamma M_e^2}\left(1 - \frac{P_0}{P_e}\right)\right)
\end{aligned}
$$

Notice that C is independent of the current true velocity of the rocket V_r. It IS however, dependent on ρ_e and P_e which are functions of altitude. Notice also, that $C \to U_e$ as $M_e \to \infty$ and/or $P_e \to P_0$.

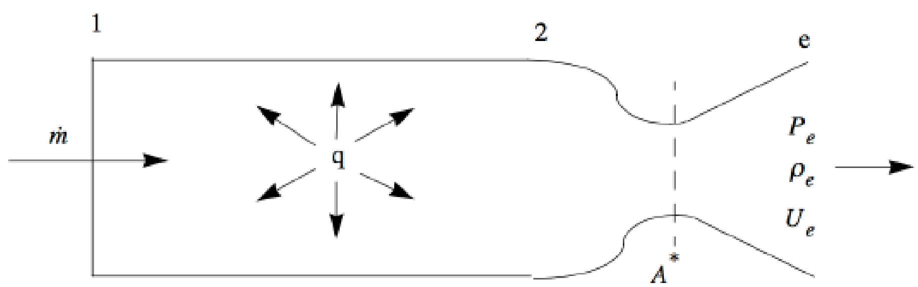


Figure 4: Stationed Diagram of a Rocket Engine

The theoretical max C, notated C_{max}, is expectedly a function of both the fuel combustion and engine geometry. If we start with the energy balance equation $h_{t2} = h_{t1} + q = h_e + \frac{1}{2}U_e^2$, and we apply the assumptions that

1. $h_{t1} \ll q$: implying heat of combustion is far greater than heat of the entering fuel.
2. $P_{t2}/P_0 \gg 1$: Nozzle pressure ratio is very large
3. $A_e/A^* \gg 1$: Nozzle Area ratio is large

We have the theoretical equation where $h_{t2} = q = \frac{1}{2}U_e^2 = \frac{1}{2}C_{max}^2$. What this yields is the following approximation chain.

$$C_{max} \approx \sqrt{2q} \approx \sqrt{2h_{t2}} \approx \sqrt{2C_p T_{t2}} \approx \sqrt{\frac{2\gamma}{\gamma - 1}\left(\frac{R_u}{M_w}\right)T_{t2}}$$

Where M_w is the molar mass of the gas in kg/mol.

Specific impulse is simply defined as thrust per unit WEIGHT of propellant flow. Hence the g_0 term. **The most important thing to remember is C fully defines I_{sp} and vice-versa.**

$$I_{sp} = \frac{I}{m g_0} = \frac{T}{\dot{m} g_0} = \frac{C}{g_0}$$

Combustion chamber pressure internals

One important tip to remember is that the critical pressure ratio required for choked flow through any orifice is.

$$\frac{P_t}{P_b} \geq \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}} \approx 1.89 \text{ for air}$$

The definitions of the pressure here are a little tricky

1. P_b is the back pressure or ambient pressure. This is a STATIC pressure.
2. P_t is the chamber pressure. This is a TOTAL pressure.

With this in mind, it is important to realize that for most rocket engines, referring to Figure 4, the ratios of $\frac{P_t}{P_b} = \frac{P_{t2}}{P_0}$ are far above the threshold values needed for choke, hence **it is very safe to assume a rocket engine's flow is always choked if started.**

The following are some key relations and their notes and assumptions for a rocket engine. Refer to Figure 4 for station values.

$$T_{t2} = T_{t1} + \frac{q}{C_p}$$

$$P_{t2} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{\sqrt{\gamma R T_t}}{\gamma A^*}\dot{m}$$

$$\frac{P_{t2}}{P_{t1}} = \left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right)\left(\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right)^{\frac{\gamma}{\gamma-1}} \xrightarrow{M_1^2 \ll 0} \left(\frac{1}{1+\gamma M_2^2}\right)\left(\frac{1+\frac{\gamma-1}{2}M_2^2}{1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{P_2}{P_1} = \left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right) \xrightarrow{M_1^2 \ll 0} \left(\frac{1}{1+\gamma M_2^2}\right)$$

1. Equation 1 essentially leverages $h_{t2} = h_{t1} + q$.
2. Equation 2 relies on the fact that the portion between station 2 and the choke point is isentropic. It is derived from conservation of mass with the assumption that you can use P_{t2} and T_{t2} values along with A^* and $f(M^* = 1) = 1$ interchangeably in the conservation of mass equation, which greatly simplifies things.
3. Equation 3 and 4 have simplified approximations that rely on the fact that $M_1^2 \ll 1$, which is reasonable since fuel entering a rocket combustion chamber is super slow.

C^* efficiency

Generally, it is hard to measure γ, R, T_t for combustion products in propulsion testing. However, it is easier to measure P_{t2} using transducers and $\dot{m}$, A^* are usually set by engineering design. Hence, if you lump many of the difficult-to-measure terms in mass conservation into a single C^* value, you get a practical definition of efficiency.

$$C^* = \frac{P_{t2}A^*}{\dot{m}} \quad ; \quad \eta_{C^*} = \frac{\frac{P_{t2}A^*}{\dot{m}}|_{measured}}{\frac{P_{t2}A^*}{\dot{m}}|_{ideal}} = \frac{P_{t2}|_{measured}}{P_{t2}|_{ideal}}$$

The Rocket Equation

Starting with a simple force balance on a rocket and integrating properly, the famous rocket equation can be derived, with drag and gravitational pull considered.

$$\begin{aligned}\Delta V_r &= \Delta V_r|_{ideal} - \Delta V_r|_{grav} - \Delta V_r|_{drag} \\ \Delta V_r|_{grav} &= \int_0^{t_b} g\sin(\theta(t))dt \\ \Delta V_r|_{drag} &= \int_0^{t_b} \left(\frac{D}{M_r(t)}\right) dt \\ \Delta V_r|_{ideal} &= C\ln\left(\frac{M_{ri}}{M_{rf}}\right)\end{aligned}$$

Where

- M_{ri} = initial mass
- M_{rf} = mass at burnout
- t_b = time of burnout
- θ = flight angle wrt horizontal (can be function of t)
- $M_r(t)$ = vehicle mass (DEFINITELY a function of t).
- D = drag (usually not a function of t).
- g = duh, (can be function of t depending on problem).
- C = effective exhaust velocity.

The remaining concepts in this chapter on rocket engines are quite niche, hence only textbook equations are screenshotted.

Drag minimization

TLDR: the more slender the rocket design, the better.

$$\Delta V_{rocket}|_{drag} \cong \frac{FrontalArea_{rocket}}{2density_{rocket}Volume_{rocket}} \int_0^{t_b} \rho C_D C^2 \left(ln\left(\frac{M_{rocket_i}}{M_{rocket}}\right)\right)^2 \left(\frac{M_{rocket_i}}{M_{rocket}}\right) dt$$
$$\sim \frac{1}{Length_{rocket}}. \qquad (7.65)$$

C_F thrust coefficient

7.10 The thrust coefficient

The thrust coefficient provides a useful dimensionless measure of engine thrust.

$$C_F = \frac{T}{P_{t2}A^*} = \frac{\dot{m}U_e + (P_e - P_0)\,A_e}{P_{t2}A^*} = \left(\frac{P_e}{P_{t2}}\right)\left(\frac{A_e}{A^*}\right)\left(\gamma {M_e}^2 + 1 - \frac{P_0}{P_e}\right) \tag{7.66}$$

This rather complicated looking expression can be written in terms of the nozzle exit Mach number and pressure

$$C_F = \frac{1}{\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}} \frac{\left(\gamma {M_e}^2 + 1 - \frac{P_0}{P_e}\right)}{M_e\left(1 + \frac{\gamma-1}{2}{M_e}^2\right)^{\frac{1}{2}}} \tag{7.67}$$

where the nozzle flow has been assumed to be isentropic. For a rocket operating in a vacuum, with a very large expansion ratio $M_e \to large$, the thrust coefficient has an upper limit of

$$C_{F_{\max}} = \frac{\gamma}{\left(\frac{\gamma-1}{2}\right)^{\frac{1}{2}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}}. \tag{7.68}$$

Chapter 8: Multistage Rockets

The multistage rocket problem is largely an optimization problem. For the purposes of these notes, the optimization aspect of it (using of Lagrange multipliers) is not explored. Additionally, only 2-stage rockets are analyzed.

Notations on Ratios ($\lambda_i, \mathcal{E}_i, R_i$)

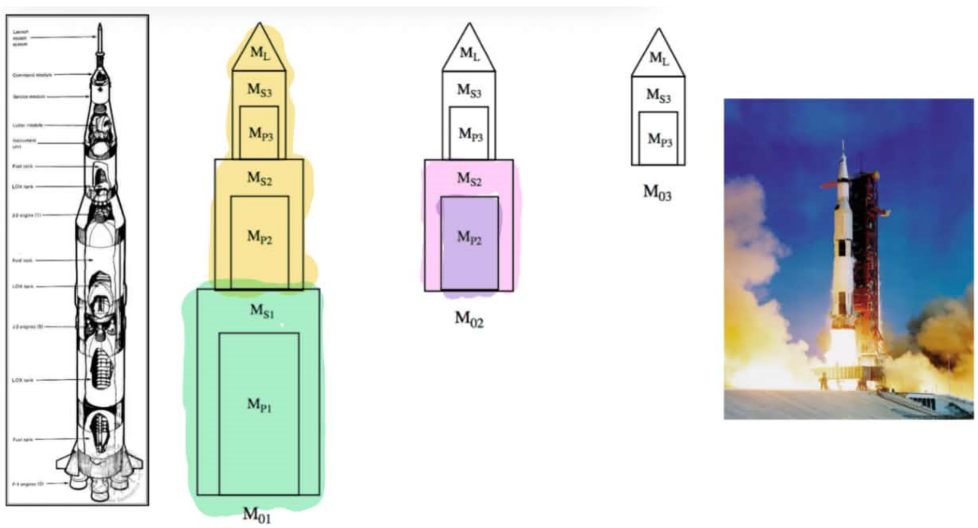


Figure 8.1: *Three stage rocket notation.*

Define the following variables.

Payload ratio $\Rightarrow$ $\dfrac{\text{"what this stage needs to carry"}}{\text{"the weight of this stage"}}$

$$\lambda_i = \frac{M_{0(i+1)}}{M_{0i} - M_{0(i+1)}} = \frac{M_{0(i+1)}}{M_{Si} + M_{Pi}}$$

$$\lambda_n = \frac{M_{0(n+1)}}{M_{0n} - M_{0(n+1)}} = \frac{M_L}{M_{0n} - M_L} \tag{8.2}$$

Structural coefficient * completely intrinsic to the i^{th} stage itself

"what fraction of my stage's mass can be structural?"

$$\varepsilon_i = \frac{M_{S_i}}{M_{0i} - M_{0(i+1)}} = \frac{M_{S_i}}{M_{S_i} + M_{P_i}} \tag{8.3}$$

Mass ratio

"Ratio of init & final masses after i^{th} stage burns"

$$R_i = \frac{M_{0i}}{M_{0i} - M_{Pi}} = \frac{1+\lambda_i}{\varepsilon_i + \lambda_i}. \tag{8.4}$$

Notice that the definition for R_i makes it very convenient to plug back into the famous Rocket Equation to yield

$$V_n = \Delta v_1 + \Delta v_2 + \cdots + \Delta v_n = \sum_{i=1}^{n} C_i \ln(R_i) = \sum_{i=1}^{n} C_i \ln(\frac{1+\lambda_i}{\mathcal{E}_i + \lambda_i})$$

Where V_n is the final velocity of the n-stage rocket and Δv_i is the change in velocity due to full burn of stage i.

The last compound ratio useful to define is Γ the Payload Fraction.

$$\begin{aligned}\Gamma = \frac{M_L}{M_{01}} &= \left(\frac{M_{02}}{M_{01}}\right)\left(\frac{M_{03}}{M_{02}}\right)\cdots\left(\frac{M_L}{M_{0n}}\right)\\ &= \left(\frac{\lambda_1}{1+\lambda_1}\right)\left(\frac{\lambda_2}{1+\lambda_2}\right)\cdots\left(\frac{\lambda_n}{1+\lambda_n}\right)\\ \ln(\Gamma) &= \sum_{i=1}^{n} \ln\left(\frac{\lambda_i}{1+\lambda_i}\right)\end{aligned}$$

Lagrange α method

Without attention to the derivation of the α Lagrange multiplier term, there are 3 slightly modified relations to the aforementioned ration notations that can be rewritten in terms of α.

$$\begin{aligned}\lambda_i &= \frac{\alpha\mathcal{E}_i}{C_i - C_i\mathcal{E}_i - \alpha}\\ V_n &= \sum C_i \ln(\frac{C_i - \alpha}{\mathcal{E}_i C_i})\\ \ln(\Gamma) &= \sum_{i=1}^{n} \ln\left(\frac{\alpha\mathcal{E}_i}{C_i - C_i\mathcal{E}_i - \alpha + \alpha\mathcal{E}_i}\right)\end{aligned}$$

Direct Method (α-less)

There is a direct method of solving for an optimal multi-stage design shown on the next page. In the long process, you can solve for $V_2 = f(\Gamma, C_1, C_2, \epsilon_1, \epsilon_2, M_{P2}/M_{P1})$. Since values of $\Gamma, C_1, C_2, \epsilon_1, \epsilon_2$ are usually given, you are really only optimizing over the propellant mass ratio. $V_2 = f(M_{P2}/M_{P1})$. Do this by setting

$$\frac{dV_2}{d(M_{P2}/M_{P1})} = \frac{df(M_{P2}/M_{P1})}{d(M_{P2}/M_{P1})} = 0$$

8.4.1 Two stage design

Consider a two stage design.

$$\begin{aligned}
&V_2 = C_1\, ln\left(\frac{M_{01}/M_{P1}}{M_{01}/M_{P1}-1}\right) + C_2\, ln\left(\frac{M_{02}/M_{P2}}{M_{02}/M_{P2}-1}\right) \\
&M_{01} = M_{S1} + M_{P1} + M_{S2} + M_{P2} + M_L \\
&M_{02} = M_{S2} + M_{P2} + M_L \\
&M_{S1} = \left(\frac{\epsilon_1}{1-\epsilon_1}\right) M_{P1} \quad M_{S2} = \left(\frac{\epsilon_2}{1-\epsilon_2}\right) M_{P2} \quad \Gamma = \frac{M_L}{M_{01}}
\end{aligned} \tag{8.27}$$

Express the payload mass in terms of propellant masses and payload fraction.

$$M_L = \left(\frac{\Gamma}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_1}\right) M_{P1} + \left(\frac{1}{1-\epsilon_2}\right) M_{P2}\right) \tag{8.28}$$

Now express the stage mass ratios in terms of the propellant mass ratio M_{P2}/M_{P1}.

$$\frac{M_{01}}{M_{P1}} = \left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_1}\right) + \left(\frac{1}{1-\epsilon_2}\right)\frac{M_{P2}}{M_{P1}}\right) \tag{8.29}$$

$$\frac{M_{02}}{M_{P2}} = \left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_2}\right) + \left(\frac{\Gamma}{1-\epsilon_1}\right)\frac{1}{M_{P2}/M_{P1}}\right) \tag{8.30}$$

Insert (8.29) and (8.30) into the first equation in (8.27). The two stage velocity increment is

f

$$\begin{aligned}
V_2 = C_1\, ln &\left(\frac{\left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_1}\right) + \left(\frac{1}{1-\epsilon_2}\right)\frac{M_{P2}}{M_{P1}}\right)}{\left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_1}\right) + \left(\frac{1}{1-\epsilon_2}\right)\frac{M_{P2}}{M_{P1}}\right) - 1}\right) + \\
C_2\, ln &\left(\frac{\left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_2}\right) + \left(\frac{\Gamma}{1-\epsilon_1}\right)\frac{1}{M_{P2}/M_{P1}}\right)}{\left(\frac{1}{1-\Gamma}\right)\left(\left(\frac{1}{1-\epsilon_2}\right) + \left(\frac{\Gamma}{1-\epsilon_1}\right)\frac{1}{M_{P2}/M_{P1}}\right) - 1}\right)
\end{aligned} \tag{8.31}$$

Chapter 9-11: Solid Rocket Fuel, Hybrid Rockets

Solid Rocket Fuel

There are 2 types of solid rocket fuel

- homogenous: oxidizer and fuel contained in the same molecule.
- heterogenous: oxidizer and fuel are separate and form a mixture. Aluminum sometimes added for increased combustion efficiency (example: Space Shuttle SRBs).

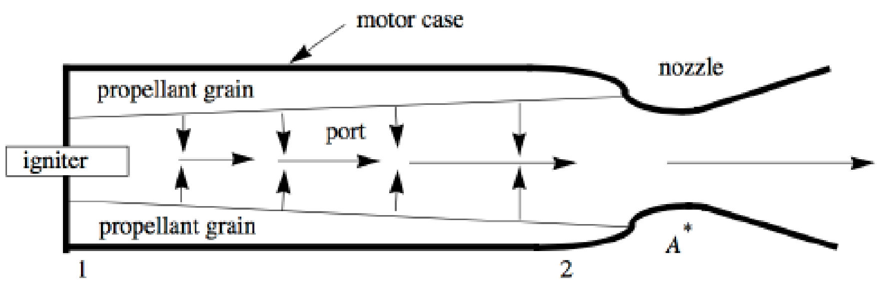
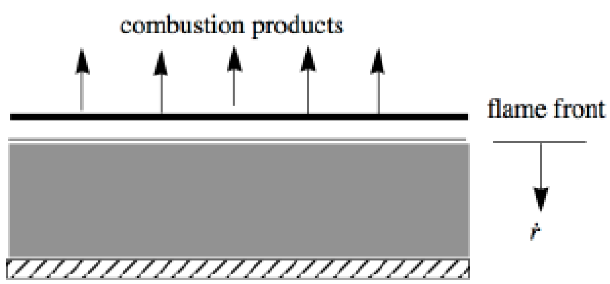


Figure 5: Schematic of a solid rocket booster



Two key first-principles equations that govern SRB performance include gas generation rate and surface regression speed.

$$\dot{m}_g = \rho_p A_b \dot{r} \quad ; \quad \dot{r} = \frac{K}{T_1 - T_p}(P_{t2})^n$$

Where

- ρ_p: solid prop density
- A_b: area of burning surface.
- $\dot{r}$: surface regression speed.
- $\dot{m}_g$: rate of gas generation at propellant surface

- P_{t2}: combustion chamber pressure
- K: **EMPIRICAL** propellant constant
- T_1: **EMPIRICAL** detonation temperature
- T_p: Propellant temperature
- n: **EMPIRICAL** exponent term

Here is a massive SRB equation dump.

$$\frac{dV}{dt} = \dot{r} A_b$$

$$P_{t2} = \left(\left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{K\left(\rho_p - \rho_g\right)}{\gamma\left(T_1 - T_p\right)} \left(\frac{A_b}{A^*} \right) \sqrt{\gamma R T_{t2}} \right)^{\frac{1}{1-n}}$$

$$\tau = \frac{\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}}{(\gamma R T_{t2})^{1/2}} \left(\frac{V}{A^*} \right)$$

$$P_{t2}(t) = \overline{P_{t2}} + p_{t2}(t)$$

$$\frac{p_{t2}}{p_{t2}(0)} = e^{-\frac{1-n}{\tau} t}$$

Let's stop here briefly, because the equation above implies that

- if $n < 1$: **STABLE**, small pressure deviations are OK since nozzle can handle more mass output than the additional gas generated.
- if $n > 1$: **UNSTABLE**, small pressure deviations are BAD since nozzle can not handle more mass output than the additional gas generated. SRB explodes.

Continuing the equation dump

$$P_{t2} = \left(\alpha \frac{r}{r_i} \right)^{\frac{1}{1-n}}$$

$$\alpha = \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{K\left(\rho_p - \rho_g\right)}{\gamma\left(T_1 - T_p\right)} \sqrt{\gamma R T_{t2}} \left(\frac{2\pi r_i L}{A^*} \right)$$

$$\frac{r}{r_i} = \left(1 + \left(\frac{1-2n}{1-n} \right) \frac{K \alpha^{\left(\frac{n}{1-n}\right)} t}{\left(T_1 - T_p\right) r_i} \right)^{\frac{1-n}{1-2n}} \qquad \text{n} \neq 0.5$$

$$\frac{r}{r_i} = e^{\frac{K \alpha t}{(T_1 - T_p) r_i}} \qquad n = 0.5$$

$$t_{\text{burnout}} = \left(\left(\frac{r_f}{r_i} \right)^{\frac{1-2n}{1-n}} - 1 \right) \left(\frac{1-n}{1-2n} \right) \tau_{\text{burn}} \qquad \text{n} \neq 0.5$$

$$t_{\text{burnout}} = \ln \left(\frac{r_f}{r_i} \right) \tau_{\text{burn}} \qquad \text{n} = 0.5$$

Hybrid Rockets

Hybrid rockets offer the main advantages of

1. being safer, since liquid oxidizer and solid fuel are stored separately, and the presence of a one-way pressure driven system makes it such that the two fuels cannot intimately mix in the case of structural failure and cause catastrophe.

2. having a generally less complex system than liquid bi-prop, since there is only one fuel in fluid form that needs to be pumped now.

3. still retaining throttle-ability which is a capability missing in pure SRBs.

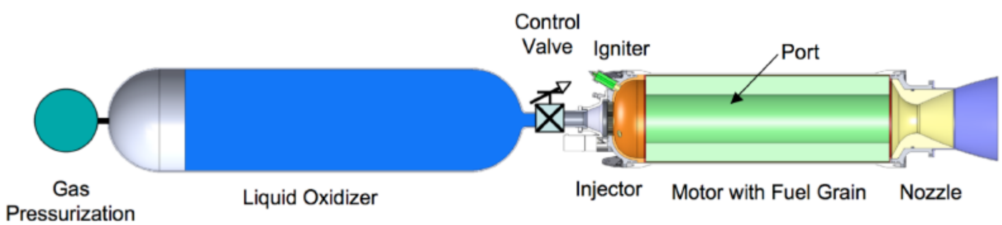
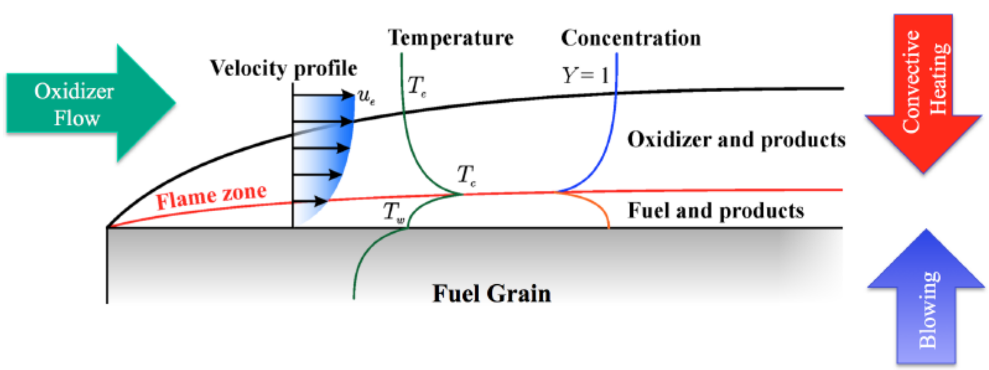


Figure 6: Hybrid Rocket Engine Schematic

Hybrid rockets typically generate hot gas using boundary-layer combustion.



Much of Hybrid Rocket Engine analysis focuses on fuel surface regression rate and mass flow through the port section. Here are some of the more fundamental equations.

$$a, n, m \rightarrow \text{empirical values for a given O and F combo, } m \text{ typically near } 0,\ 0.4 < n < 0.7$$

$$\dot{m}_{port} = \dot{m}_{ox} + \dot{m}_{fuel}$$
$$G = \frac{\dot{m}_{port}}{\pi r^2}$$
$$a(n,m) = \frac{L^{2n+m+1}}{Mass^n Time^{1-n}}$$
$$\dot{r} = aG_{ox}^n = a\left(\frac{\dot{m}_{ox}}{\pi r^2}\right)^2$$
$$OF = \frac{\dot{m}_{ox}}{\dot{m}_f}$$
$$a_o = a(1 + 1/(OF(x,t)))^n$$

Solving the pair of coupled PDEs **for the case** $n = 1/2$ yields.

$$r(x,t) = \left(r(x,0)^2 + \frac{2a}{x^m \pi^{1/2}}\left(\int_0^t \dot{m}_{ox}\left(t'\right)^{1/2} dt' + \frac{\pi^{1/2} a \rho_f x^{1-m} t}{1-m}\right)\right)^{1/2}$$
$$\dot{m}_{\text{port}}\,(x,t) = \left(\dot{m}_{ox}(t)^{1/2} + \frac{\pi^{1/2} a \rho_f x^{1-m}}{1-m}\right)^2$$

Lastly, here are some equations that I have no idea what they mean, but they are just here.

$$\dot{m}_{ox} = \frac{1}{1+f}\frac{P_c(\pi r_{initial}^2)}{C_3^2 C_2^2}$$

$$\boxed{f = 2(1+f)^{1/2}\eta + (1+f)\eta^2} \qquad \boxed{\eta = C\frac{a\rho_f}{P_c^{1/2}}}$$

$$C = C_1 C_2 C_3 = \frac{L}{r_{initial}}\frac{r_{initial}}{r^*}\frac{1}{\gamma^{1/2}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{4(\gamma-1)}}(\gamma R T_c)^{1/4} \qquad C = 1.55 \times 10^3\ (\text{m/sec})^{1/2}$$

Some key downsides to hybrid rockets include

1. low burning rates (or fuel regression rates) cause the need for multi-port design, which is more complex.
2. there is poor structural integrity for the solid fuel grain as liquid passes over.
3. length of the port is an optimization problem, so as to not have any wasted fuel OR oxidizer by the outlet of the port. The Stoichiometry is a geometry problem.
4. O/F ratio shifts are annoying to deal with
5. Refueling is a difficult process.

Addendum: Scramjets

A scramjet is very comparable to a ramjet with the following primary differences

- The inlet design is more intricate (see Busemann streamtraced inlet)
- The inlet is no longer assumed to be lossless/isentropic ($k = \frac{P_{t2}}{P_{t0}}$ pressure-recovery ratio).
- There is an additional zone, the isolator, that sits between the inlet and the combuster.
- The nozzle is purely diverging, and is where all the thrust is generated. There is no need for any converging portion of a nozzle because flow is already supersonic everywhere in the scramjet.

Isolators

The only aspect of a scramjet that deserves some additional attention is the isolator. The isolator serves 2 main reciprocal purposes

1. dampen any disturbances in the inlet from reaching the combustor
2. more importantly: protect the combustor from disturbing and potentially unstarting the inlet via large precombustion pressure rises.

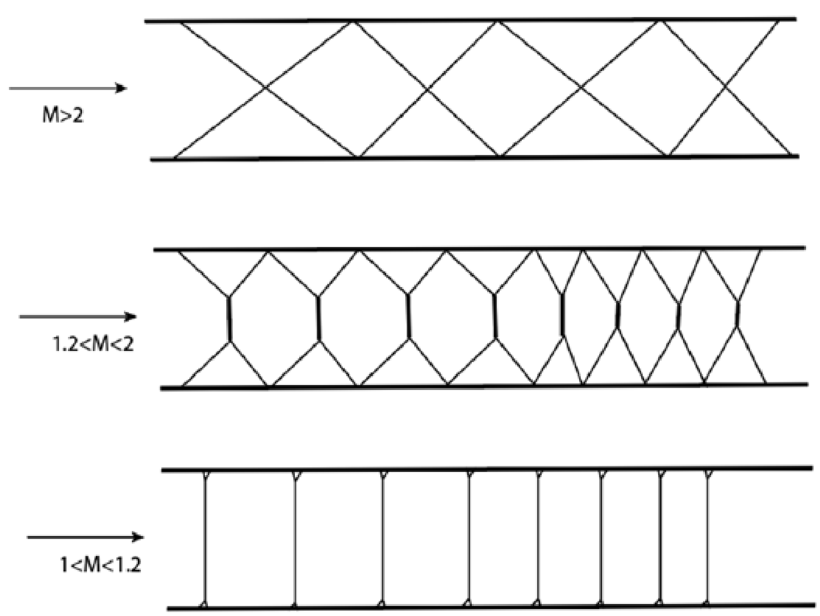


Figure 7: Isolator Types

Some cool engines and planes to know

1. **Lockheed SR-71 Blackbird**
   - *Engine:* 2 × Pratt & Whitney J58 (JT11D-20) continuous-bleed turbojets (functioning as partial ramjets at high Mach numbers).
2. **Boeing 747**
   - *Engine:* 4 × Pratt & Whitney JT9D, Rolls-Royce RB211, or General Electric CF6 high-bypass turbofans (depending on the variant; later models used the GE CF6-80 or GEnx).
3. **Concorde**
   - *Engine:* 4 × Rolls-Royce/Snecma Olympus 593 turbojets with reheat (afterburners).
4. **McDonnell Douglas F-15 Eagle**
   - *Engine:* 2 × Pratt & Whitney F100-PW-100 or -220 afterburning turbofans.
5. **Convair XF-92**
   - *Engine:* 1 × Allison J33-A-29 turbojet with an afterburner.
6. **Northrop B-2 Spirit**
   - *Engine:* 4 × General Electric F118-GE-100 non-afterburning turbofans (buried deep within the wing to minimize radar and infrared signatures).
7. **Grumman X-29**
   - *Engine:* 1 × General Electric F404-GE-400 afterburning turbofan.
8. **Republic XF-84H "Thunderscreech"**
   - *Engine:* 1 × Allison XT40-A-1 dual-unit turboprop engine driving an Aeroproducts supersonic, three-bladed propeller.
9. **NASA X-43A**
   - *Engine:* 1 × Integrated air-breathing Supersonic Combustion Ramjet (Scramjet).